Tokenize

one hour → one token

one hour t: 84 variables

$\mathbf{x}_{i,t}$ ...

$\mathbf{m}_{i,t}$ ...

$[\mathbf{x}_{i,t}; \mathbf{m}_{i,t}] \in \mathbb{R}^{168}$

MLP 168 → 256 → 64

$\mathbf{z}_{i,t} \in \mathbb{R}^{64}$

+ sinusoidal time encoding

→ one of 24 hourly tokens

Encode

24 hourly slots · mask empty hours

valid | empty hour

Transformer encoder

self-attention · 4 heads

FFN = 256

× 2 layers

pre-LN · d_model = 64

$\mathbf{H}_i \in \mathbb{R}^{24\times64}$

empty → no attention, no SSL loss

Outputs

pretraining vs downstream readout

$\mathbf{H}_i \in \mathbb{R}^{24\times64}$

SSL pretraining

before downstream pooling

... → objective

MAE · JEPA

Contrastive (pooled views)

Downstream head

mean pooling over valid hours

... mean → $\mathbf{r}_i \in \mathbb{R}^{64}$

task head